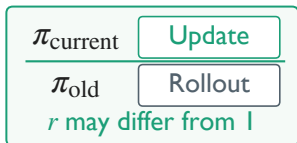
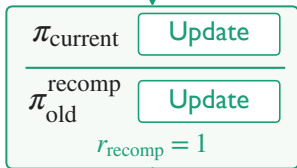


Rollout / policy-update forward
First minibatch, identical inputs and weights

AReal

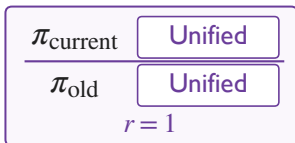


Recompute old π
Extra forward pass



PPO / GRPO objective

AReal-TIK



Direct reuse
No extra forward

$r = 1$

PPO / GRPO objective